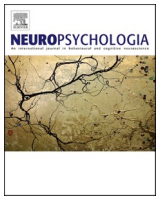
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The impact of degraded vision on emotional perception of audiovisual stimuli: An event-related potential study

Yuchen Li [a,b,d], Jing Wang [a,d], Junyu Liang [a,c,d], Chuanlin Zhu [e,1,**], Zhao Zhang [a,b,d,*,1], Wenbo Luo [a,d,***,1]

[a] Research Center of Brain and Cognitive Neuroscience, Liaoning Normal University, Dalian, 116029, China
[b] Institute of Psychology, Shandong Second Medical University, Weifang, 216053, China
[c] School of Psychology, South China Normal University, Guangzhou, 510631, China
[d] Key Laboratory of Brain and Cognitive Neuroscience, Dalian, 116029, China
[e] School of Educational Science, Yangzhou University, Yangzhou, 225002, China

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ABSTRACT

Emotion recognition will be challenged for individuals when visual signals are degraded in real-life scenarios. Recently, researchers have conducted many studies on the distinct neural activity between clear and degraded audiovisual stimuli. These findings addressed the "how" question, but the precise stage of the distinct activity that occurred remains unknown. Therefore, it is crucial to use event-related potential (ERP) to explore the "when" question, just the time course of the neural activity of degraded audiovisual stimuli. In the present research, we established two conditions: clear auditory + degraded visual (AcVd) and clear auditory + clear visual (AcVc) multisensory conditions. We enlisted 31 participants to evaluate the emotional valence of audiovisual stimuli. The resulting data were analyzed using ERP in time domains and Microstate analysis. Current results suggest that degraded vision impairs the early-stage processing of audiovisual stimuli, with the superior parietal lobule (SPL) regulating audiovisual processing in a top-down fashion. Additionally, our findings indicate that negative and positive stimuli elicit greater EPN compared to neutral stimuli, pointing towards a subjective motivation-related attentional regulation. To sum up, in the early stage of emotional audiovisual processing, the degraded visual signal affected the perception of the physical attributes of audiovisual stimuli and had a further influence on emotion extraction processing, leading to the different regulation of top-down attention resources in the later stage.

1. Introduction

The perception of emotional information is commonly reliant on the integration of multiple sensory modalities. Facial expressions, body movement, and vocal prosody are all examples of sensory inputs that can be integrated into a unified signal within the brain, facilitating emotional perception (Stein et al., 1993). However, the accurate perception of emotional information becomes particularly challenging when stimulus information is degraded, as observed in studies involving limited visual input (de Boer et al., 2021a; de Boer et al., 2021b). Research has demonstrated that visual degradation can substantially impact emotional perception (Colijn et al., 2017). This is particularly evident in older adults suffering from age-related macular degeneration (AMD), where impaired vision may lead to distorted or blurred facial details, ultimately culminating in impaired emotional recognition (Johnson et al., 2017) and social interaction (Lane et al., 2018). Therefore, it is crucial to develop a more comprehensive understanding of the impact of degraded vision on the multisensory perception of emotional information.

In recent years, researchers have increasingly focused on investigating these questions using a combination of behavioral and neuroimaging methods. De Boer et al. (2021a, 2021b) have reported that

* Corresponding author. Research Center of Brain and Cognitive Neuroscience, Liaoning Normal University, Dalian, 116029, China.
** Corresponding author.School of Educational Science, Yangzhou University, Yangzhou, 225002, China
*** Corresponding author. Research Center of Brain and Cognitive Neuroscience, Liaoning Normal University, Dalian, 116029, China.
E-mail addresses: psyclzhu@yzu.edu.cn (C. Zhu), zz543887294@163.com (Z. Zhang), luowb@lnnu.edu.cn (W. Luo).
[1] Corresponding author: These authors contributed equally to the work.

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degraded visual signals have a greater impact on emotional recognition than degraded auditory signals. Participants were asked to complete an emotion recognition task across eight conditions, including degraded visual (Vd), degraded auditory (Ad), clear visual (Vc), clear auditory (Ac), clear auditory + degraded visual (AcVd), degraded auditory + degraded visual (AdVd), clear visual + clear auditory (AcVc), and clear visual + degraded auditory (AdVc). Behavioral results indicated that accuracy was lower in the AcVd and AdVd conditions than in the AcVc condition, but not in the AdVc condition when compared with the baseline AcVc condition. De Boer et al. (2021a, 2021b) proposed that the degradation of visual stimuli impairs the emotional recognition accuracy of audiovisual (AV) stimuli. Overall, the aforementioned studies provide evidence for the influence of visual features on emotional recognition of AV stimuli from a behavioral standpoint.

Furthermore, there has been considerable interest among researchers in examining the neural mechanisms underlying both clear and degraded AV stimuli (Bhat et al., 2015; Regenbogen et al., 2018). Regenbogen et al. (2018) conducted an fMRI study that employed two signal-to-noise ratio (SNR) conditions to investigate the impact of degraded and clear stimuli on brain activation. The results indicated that the intraparietal sulcus (IPS) was activated during the processing of the degraded AV stimuli (AdVd). Degraded information requires greater attentional resources to extract effective signals (White and Langdon, 2021), whereas the IPS may serve to protect ongoing tasks from extraneous information (Blanco-Elorrieta et al., 2020). To investigate the modulation of visual and auditory cortical connectivity by IPS at different levels of signal clarity. Regenbogen et al. (2018) proposed a Dynamic Causal Modeling (DCM). It suggests that the processing of hard-to-perceive AV stimuli requires top-down regulation of the IPS on the connection between primary visual and auditory cortices, while the processing of easy-to-perceive AV stimuli is independent of the IPS and occurs directly via the connection between these cortices.

The findings of Regenbogen et al. (2018) highlighted the significance of the clarity of signals in cognitive processing. However, how signal clarity affects the time course of emotional perception during the processing of emotional AV stimuli remains unknown. Collignon et al. (2008) adopted the matching AV stimuli as either emotional congruent (i.e., fearful face and fearful voice) or incongruent (i.e., fearful face and disgust voice) emotions. The sensory signal contained clear and degraded visual stimuli, auditory stimuli, and AcVc and AcVd stimuli. In the incongruent condition, they found that the participants tended to report the emotion of the AV stimuli to the emotion of visual modality when visual signals were clear and to report the emotion of the AV stimuli to the emotion of auditory when visual signals were noise. It means that the advantages of the visual modality in emotional perception are not absolute and unalterable (Ernst and Bulthoff, 2004). In response to this concern, the maximum likelihood estimation model (MLE) may be a potential solution (Deneve and Pouget, 2004; Kersten and Yuille, 2003), which proposes a dynamic weighting approach and takes into account the reliability of sensory modalities (Shaikh, 2022). For emotional AV processing, emotions may affect the brain's judgment of the reliability of the signal because emotional stimuli require fewer attention resources (Carstensen and DeLiema, 2018; Neuberg et al., 2011). In addition, emotion processing has changed over time (Luo et al., 2010). It means that the existing research is unable to provide a definitive answer as to when the degraded visual signal will affect the emotional AV stimuli. Therefore, further research is necessary to provide new evidence.

The evidence indicating distinct neural mechanisms between degraded and clear auditory-visual stimuli substantiates the existence of feedback regulation from higher brain regions responsible for integrating processing into the primary auditory and visual cortex (Chai et al., 2021). Despite previous studies offering theoretical models to account for these findings and addressing the "how" question, the precise stage of processing during which this regulation occurs remains unknown, thus raising the question of "when". For example, it remains unclear whether the regulation of higher brain areas in response to emotional auditory-visual stimuli takes place during the stage of emotional information encoding or the higher-level cognitive evaluation stage. Currently, these inquiries remain unanswered. In the current study, we aim to explore the time course of how degraded visual signals affect the emotional perception of AV stimuli by analyzing event-related potentials (ERPs) and Microstate analysis. These methods allowed us to detect dynamic variations between conditions at a given time and measure them using statistical data on microstate activation. Specifically, we manipulated the signal-to-noise ratio (SNR) of facial emotion pictures to produce clear (AcVc) and blurred (AcVd) visual stimuli. Drawing from previous findings (Doi and Shinohara, 2015), we presented participants with audio-visual stimuli for three different emotions, comprising positive (happy), neutral, and negative (fear) emotions, and requested that they assess the emotional valence based on both auditory and visual cues.

We selected the well-established event-related potential (ERP) components and examined them using traditional mean amplitude ERP analysis. Building upon our previous research, we proposed a three-stage model for the processing of AV stimuli, representing a consistent evaluation of "affective meaning", and the P1, N170, and LPC (late positive component) related to emotional processing were reviewed (Liang et al., 2022). The first stage was characterized by auditory and visual cortex signal extraction and encoding, as well as fine processing. De Gelder and Vroomen (2000) proposed that in the early stage of AV processing, one sensory input could enhance the perceptual processing of another. For instance, emotional sounds could promote visual stimuli processing by activating the occipital lobe and fostering early visual P1 components (Brosch et al., 2009). On the other hand, the primary auditory cortex also responded to visual inputs (Romei et al., 2007), measured by N1 and P2 components (Balconi and Carrera, 2011; Pourtois et al., 2000). Therefore, selecting electrodes at frontocentral and occipitotemporal clusters proved beneficial in monitoring variability in visual and auditory ERP components, respectively. Prior research showed that emotional AV stimuli produced a larger P1 than neutral stimuli (Brosch et al., 2009) and were influenced by negative emotions (Stefanou et al., 2019). The second stage was during N170 components, which was mainly influenced by attention bias of negative emotions (Luo et al., 2010) and sensitivity to frightful AV stimuli (Stefanou et al., 2019). These studies corroborated our emphasis on these ERP components, reflecting attention bias and emotional encoding of the primary visual and auditory cortex. The final stage involved the evaluation of affective meaning. LPC was the representative index of the high-level stage. According to the task requirements, the cognitive resources were distributed by the top-down regulation mechanism (Frühholz et al., 2011). There was another important component that had not been addressed in our former work: early posterior negativity (EPN), which was an indicator of the priority processing of emotional stimuli. Prior research has demonstrated that in both low- and high-contrast tasks, unpleasant pictures lead to an increase in EPN when compared to neutral images (Schindler and Bublatzky, 2020). Additionally, Schupp et al. (2007, 2008) suggested that EPN is sensitive to visual stimulus characteristics, which allows for top-down emotional evaluation. Lee et al. (2010) discovered that, in the noise-alone task (without facial stimuli presented), a larger EPN is consistently activated when participants report sighting a fearful face. Based on these findings, we chose to focus on the P1, N170, and EPN components elicited by occipitotemporal electrodes and the N1, P2, and P3 components induced by the frontocentral electrode. We hypothesized that visual clarity would impact AV processing throughout the time course, and additionally, we expected variations among the three emotional AV stimuli that were modulated by visual clarity.

In order to further observe the dynamic variations of brain function during the processing of AV stimuli between conditions, we employed microstate analysis to differentiate and compare the topological composition of scalp ERP while also tracking fluctuations in brain

functional states as previously outlined (Koenig et al., 2014; Murray et al., 2008). Microstate analysis is used to identify the microstates in ERP data for each condition. And the assignment can be used to identify differences in the timing of a given microstate class (i.e., onset, offset, and duration), which is a very elegant and efficient way to exploit the information yielded by the high temporal resolution of the ERP data (Koenig et al., 2014). In our previous research, we found that there was a different microstate class for the two experiments during the time window of the LPC component (Liang et al., 2022). The findings of microstate analysis provided more details for the neural mechanisms of emotional AV processing, making a complement to the ERP findings. In the current study, we predicted that during the early stages of processing, the microstate patterns would exhibit significant differences between the clear and degraded conditions. This is because clear and degraded visual signals rely differently on the primary visual cortex, leading to a reweighting effect (Deneve and Pouget, 2004; Kersten and Yuille, 2003; Shaikh, 2022). During later stages, the microstate patterns should become more uniform, as top-down attentional regulation from higher brain regions separately completes the emotional recognition of clear and degraded AV stimuli. Nevertheless, the strength and duration of activation would remain influenced by emotional valence and visual clarity.

2. Materials and methods

2.1. Participants

A total of 31 college students participated in this study, comprising 15 males and 16 females with an average age of 21.58 ±0 .44 years. All participants were right-handed and had normal or corrected-to-normal vision. Before the experiment, participants were informed about the study's objectives and provided their consent. Participation in the study was voluntary, and the participants were guaranteed compensation upon completion. The Ethics Committee of Liaoning Normal University approved the study, which adhered to the Declaration of Helsinki. G*Power 3.1 was used to determine the sample size, with a total of 19 participants selected ($f = 0.25$, $\alpha = 0.05$, and $1 - \beta = 0.8$). Based on Cohen's (1992) effect size standard, the study had sufficient statistical power for testing (Cohen, 1992). The final analysis excluded two participants due to a trial rejection rate greater than 20% after artifact rejection.

2.2. Materials

Before the formal experiment, 20 participants were recruited to evaluate the emotional images and sounds' valence and arousal levels. Before the experiment, participants were informed about the study's objectives and provided their consent. Participants not involved in the formal experiment rated the emotional valence on a scale of 1–9, where 1 indicated "extremely positive" and 9 indicated "extremely negative". Similarly, emotional arousal was rated on a scale of 1–9, with 1 indicating "extremely calm" and 9 indicating "extremely excited". Table 1 displays the mean valence and arousal levels of the emotional stimuli.

Thirty images were selected from the Chinese Facial Affective Picture System (Gong et al., 2011) and were equally categorized into neutral, happy, and fearful faces, with each category containing ten images. The images of facial expressions had a measured view angle of 4.1 × 5.3°. Analysis showed a significant main effect of valence, $F(2, 27) = 400.06$, $p < .001$, $\eta_p^2 = 0.978$. Significantly higher valence ratings were observed for fearful and happy facial expressions compared to neutral facial expressions (all p values $< .001$). Fearful facial expressions also had significantly higher valence ratings compared to happy facial expressions ($p < .001$). Similarly, a significant main effect of arousal was observed, $F(2, 27) = 431.34$, $p < .001$, $\eta_p^2 = 0.980$. Both fearful and happy facial expressions had significantly higher arousal ratings than neutral facial expressions (all p values $< .001$). However, no significant difference was detected between the arousal ratings for fearful and happy facial expressions ($p > .05$).

Table 1
The mean valence and arousal of emotional stimulus.

Type	Emotion	Valence	Arousal
Pictures	negative	7.26 ± .14	7.21 ± .09
	neutral	5.04 ± .04	3.19 ± .06
	positive	2.81 ± .11	6.88 ± .14
Voices	negative	6.90 ± .23	6.84 ± .10
	neutral	4.55 ± .05	3.17 ± .10
	positive	2.71 ± .18	6.92 ± .37

The auditory stimuli employed in this study were selected from the Montreal Affective Voices database (Belin et al., 2008) and equally divided into three categories: neutral, fearful, and happy vocalizations (n = 4). Before the presentation, the auditory stimuli underwent normalization to 90% of their maximum amplitude and digitization at a sampling rate of 44.1 kHz with 16-bit resolution. The original recordings were edited to an approximate length of 735ms (M = 734ms, SD = 49ms), with the extraction and correction of emotional information ensuring consistency in the physical properties of the sounds. Dual-channel headphones were used to present the stimuli at approximately 80 dB, and the validity of sound emotion was confirmed through additional ratings. The analysis demonstrated a significant main effect of valence ($F(2, 6) = 223.51$, $p < .001$, $\eta_p^2 = 0.987$), with both fearful and happy stimuli engendering greater responses than neutral stimuli (all p values = .002). Fearful stimuli generated greater responses than happy stimuli ($p < .001$). Additionally, a significant main effect of arousal was observed ($F(2, 6) = 126.58$, $p < .001$, $\eta_p^2 = 0.977$), with both fearful and happy stimuli eliciting greater responses than neutral stimuli ($p < .001$ and $p = .001$, respectively). No significant difference was found between fearful and happy stimuli (p = .826). Finally, the AV stimuli were designed to ensure consistency between the emotional valence and gender of both vocalizations and faces.

The study consisted of two visual stimulus conditions: clear and degraded. Six levels of visual signal were presented, varying from 10% clear (90% degraded) to 60% clear (40% degraded) with a 10% gradient unit of visual noise. The visual noise was generated using MATLAB by adding salt and pepper noise. The clear condition was 100% clear, while the degraded condition ranged from 10% to 60% clear. Thirty individuals who did not participate in the formal experiment were recruited to rate discrimination of emotional images. Using SPSS Statistics (version 20.0; IBM, Somers, USA), response accuracy was analyzed using a two-factor repeated-measures ANOVA with clarity (clear and degraded) and emotion (neutral, positive, and negative) as within-subject factors. Results revealed a significant effect of visual clarity on response accuracy ($F(1, 29) = 191.56$, $p < .001$, $\eta_p^2 = 0.869$). Clear stimuli ($M \pm SE$, 95.70 ± 0.72%) resulted in greater accuracy than degraded stimuli (87.10 ± 0.70%). No significant main effect of emotion or interaction effect between clarity and emotion was found (all p values $> .05$). These findings suggest that while judgment accuracy of degraded facial expressions was significantly lower than that of clear faces, it remained much higher than chance level. The facial stimuli utilized in the study are presented in Fig. 1.

2.3. Procedure

The experiment was conducted in a quiet room in a controlled environment. The background color for the presentation of all experimental stimuli was black, and all stimuli were presented on a 17″ LED monitor computer screen with a refresh rate of 60Hz and a spatial resolution of 1024 × 768. The viewing distance was maintained at 70 cm.

The experimental design utilized a 2 (Clarity: clear and degraded) × 3 (Emotion: neutral, positive, and negative) factorial design. Each trial

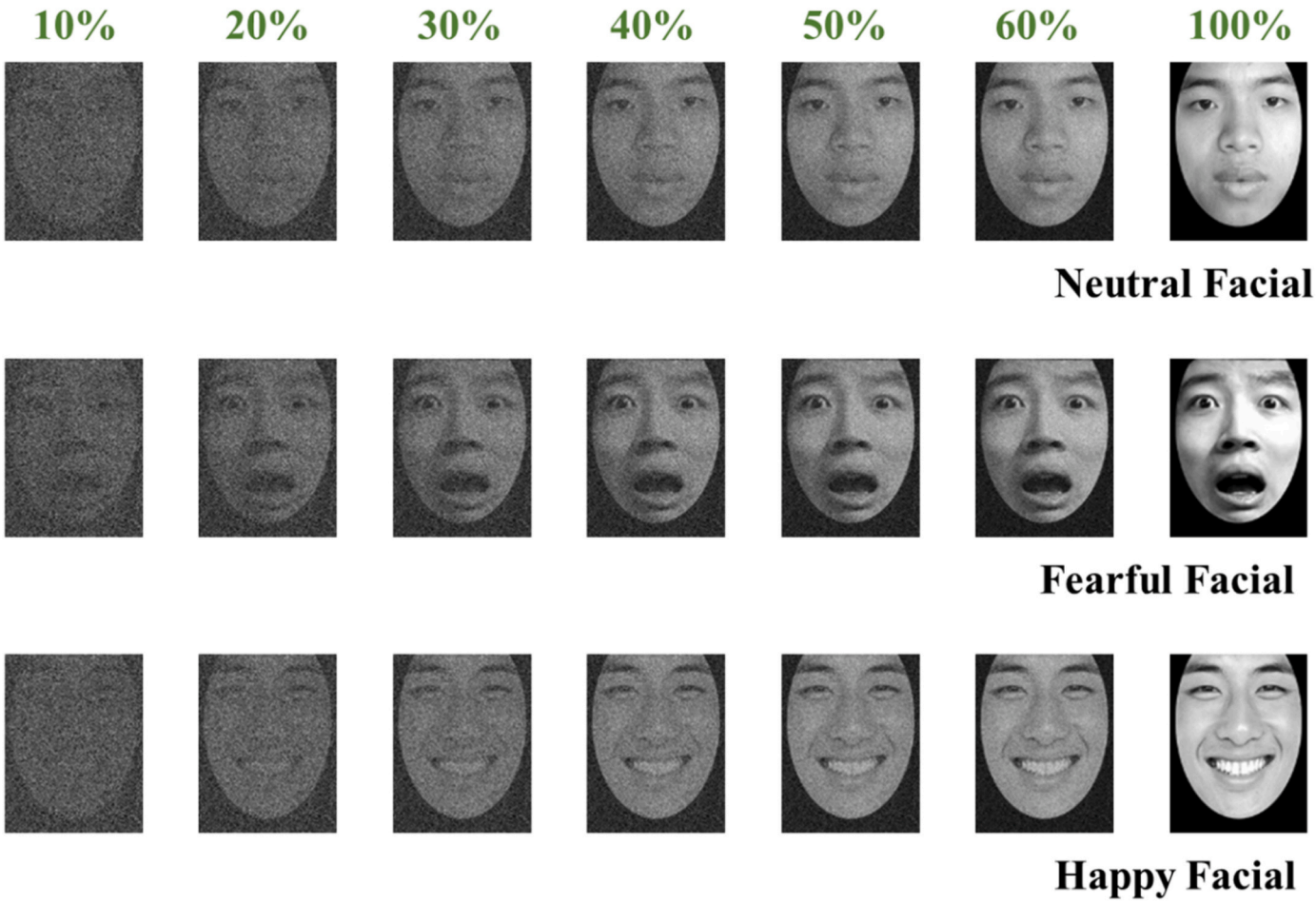


Fig. 1. The facial stimuli used in present study, including clear and degraded conditions. The adjustment of the degraded visual signal included six levels, from 10% clear (90% degraded) to 60% clear (40% degraded) with a 10% gradient unit as visual noise. The clear condition was 100% clear, and the degraded was 10%–60% clear. The facial expression induced negative, neutral, and positive emotion, respectively.

commenced with the presentation of a fixation cross for 500–700ms, followed by emotional AV stimuli lasting for 800ms (including AcVc and AcVd). After the stimuli disappeared, a period of blank screen then appeared for participants to complete the judgment task with unlimited time. They were instructed to respond to the emotional valence of the AV stimuli as quickly and accurately as possible by pressing keys encompassed by these keyboard specifications: "F" for negative emotion, "J" for neutral emotion, and the spacebar for positive emotion. The study comprised a practice phase (24 trials) and a test phase (360 trials, partitioned into five blocks). The participants were directed to provide prompt and precise responses to the emotional valence of the stimuli.

2.4. Electroencephalogram (EEG) recordings

A 64Ag/AgCI electrode EEG system (Brain Products ERP, Germany) was used to record electroencephalogram (EEG) data, according to the International Standard 10–20 system. The EEG signals were continuously sampled at a rate of 500Hz/channel, and a bandpass filter of 0.01–250Hz was applied. The left mastoid was used as the online reference, and a reference electrode standardization technique (REST) (Yao, 2001) was used as the offline reference. REST was a method that was proposed to approximately standardize the reference of scalp EEG recording to a point at infinity, far from all the possible neural sources. The impedance of each electrode point was maintained below 5kΩ. Offline data analysis was performed using MATLAB R2021a (MathWorks, Natick, MA, USA) and the EEGLAB 2020 toolbox. The raw data were digitally filtered at 0.05–50Hz (24dB/oct). The EEG segment was selected as −200 to 800ms, and −200 to 0ms was selected as the baseline. The correction of artifacts due to horizontal eye movement was performed by classifying components with independent component analysis (ICA). Independent components related to horizontal eye movements were identified by a large contribution to "Eye" and distribution on the frontal scalp and were excluded. For ERP data, the baseline was set as −200 to 0ms. Epochs with simple voltage threshold deflections exceeding ±50 μV were excluded, and at least 80% of trials were retained for each condition.

2.5. Data analysis

The ERP waveform was time-locked to the onset of stimulation. Following a previous ERP study (Doi and Shinohara, 2015), the average amplitude of the auditory N1 (140–180ms), auditory P2 (194–234ms), and P3 (286–386ms) components were selected from a frontocentral cluster (F1/Fz/F2/FC1/FCz/FC2/C1/Cz/C2). Additionally, the average amplitude of the visual P1 (140–180ms), N170 (194–234ms), and EPN (286–386ms) components were selected from an occipitotemporal cluster (P7/PO7/P8/PO8).

Behavioral and ERP data were analyzed using a two-factor repeated-measures ANOVA: 2 (Clarity: clear and degraded) × 3 (Emotion: neutral, positive, and negative) in SPSS Statistics (version 20.0; IBM, Somers, USA). The mean reaction times (RTs) and mean accuracy were statistically analyzed as dependent variables. The RTs were selected as the correct response and limited between 100 and 1000ms. *P*-values were

corrected using the Bonferroni correction. Additionally, η_p^2 was calculated as a measure of effect size.

2.6. Microstate analysis

To conduct a topographic consistency test (TCT) analysis and microstate analysis, we utilized the Randomization Graphical User Interface (RAGU) developed by Habermann et al. (2018). To identify the optimal number of microstate prototype maps, the proposed cross validation procedure was applied 250 times. Each time, 15 people were randomly selected from 31 subjects as the training set, and the rest were the learning data set, testing between 3 and 10 microstate classes. Every microstate identification run used the K-means algorithm with 50 random initializations each (Koenig et al., 2014). These microstate prototypes were then fit to the grand averaged data of the six conditions (Lehmann et al., 1987). Statistical analysis of microstate parameters was carried out through the use of randomization statistics, which compared differences in microstate features between actual data sets against distributions under the null hypothesis as outlined by Koenig and Melie--García (2010). Further details regarding our TCT analysis can be found in the Appendix.

3. Results

3.1. Behavioral data

Visual clarity had a significant main effect on RTs [F (1, 30) = 18.91, $p < .001$, $\eta_p^2 = 0.387$], with clear stimuli leading to shorter RTs (359.34 ± 15.09ms) compared to degraded stimuli (375.08 ± 15.34ms). Moreover, the main effect of emotion was significant [F (2, 60) = 13.21, $p < .001$, $\eta_p^2 = 0.306$], with both negative and positive emotions resulting in significantly longer RTs (368.23 ± 15.50ms and 381.59 ± 14.84ms, respectively) compared to neutral emotions (351.81 ± 16.06ms, $p = .008$ and $p < .001$). However, the interaction effect of clarity by emotion was not significant ($p > .05$).

Visual clarity significantly affected response accuracy [F (1, 30) = 16.30, $p < .001$, $\eta_p^2 = 0.352$], with clear stimuli resulting in greater accuracy (97.65 ±0 .38%) compared to degraded stimuli (96.38 ±0 .57%). However, both the main effect of emotion and the interaction effect of clarity by emotion were not significant (all p values $> .05$).

3.2. ERP data

The mean average value of ERPs waveforms for each condition was determined and recorded in Table 2 and Fig. 2. Furthermore, Table 3 displays the main effect and interaction of each variable on the mean average value of the ERP waveforms.

Table 2
The mean average value of ERP waveforms induced by emotional audiovisual stimulus.

Clarity	Emotion	N1	P2	P3	P1	N170	EPN
AcVc	negative	−2.06 ± .23	2.04 ± .29	−1.91 ± .36	2.88 ± .32	−2.67 ± .40	3.05 ± .43
	neutral	−1.96 ± .21	1.47 ± .26	−2.21 ± .32	2.60 ± .32	−1.91 ± .34	3.19 ± .41
	positive	−1.90 ± .24	1.63 ± .31	−1.55 ± .34	2.66 ± .34	−2.12 ± .41	2.40 ± .42
AcVd	negative	−1.99 ± .23	−.07 ± .30	−2.51 ± .33	2.81 ± .31	.52 ± .40	3.77 ± .44
	neutral	−2.29 ± .23	−.15 ± .27	−3.12 ± .35	3.21 ± .31	.65 ± .38	4.56 ± .50
	positive	−1.88 ± .21	.03 ± .29	−2.17 ± .33	2.81 ± .26	.53 ± .40	3.43 ± .42

3.2.1. Frontocentral cluster

N1 The main effect of emotion was significant [F (2, 60) = 4.58, $p = .014$, $\eta_p^2 = 0.133$]; N1 in neutral emotions was larger than that in positive emotions ($p = .020$), no other Bonferroni post hoc comparisons were significant (all p values $> .05$). The main effect of clarity and the interaction effect of clarity by emotion were not significant, $p > .05$ (Fig. 2, Tables 2 and 3).

P2 The main effect of clarity was significant [F (1, 30) = 60.12, $p < .001$, $\eta_p^2 = 0.667$]; clear stimuli led to a larger P2 than degraded stimuli. The main effect of emotion was significant [F (2, 60) = 6.24, $p = .003$, $\eta_p^2 = 0.172$]. P2 was larger in negative than in neutral ($p = .008$), no other Bonferroni post hoc comparisons were significant (all p values $> .05$). The interaction between clarity and emotion was significant [F (2, 60) = 4.61, $p = .014$, $\eta_p^2 = 0.133$]. At the clear level, the auditory P2 in negative was larger than in neutral and positive stimuli (all $p < .01$), and there was no significant difference between neutral and positive stimuli ($p > .05$). At the degraded level, the main effect of emotion was not significant, $p > .05$ (Fig. 2, Tables 2 and 3).

P3 The main effect of clarity was significant [F (1, 30) = 12.37, $p = .001$, $\eta_p^2 = 0.292$]; clear stimuli elicited a larger P3 amplitude than degraded stimuli. The main effect of emotion was significant [F (2, 60) = 24.80, $p < .001$, $\eta_p^2 = 453$], P3 was larger in positive than in negative and neutral emotions ($p = .004$ and $p < .001$); also larger in negative than in neutral emotions ($p = .002$). The interaction effect of clarity by emotion was not significant, $p > .05$ (Fig. 2, Tables 2 and 3).

3.2.2. Occipitotemporal cluster

P1 The interaction between clarity and emotion was significant [F (2, 60) = 5.89, $p = .005$, $\eta_p^2 = 0.164$]. The further post hoc by using Bonferroni correction found that P1 was larger in neutral emotions than in negative and positive emotions (all p values = .05) at the degraded level, but there was no significant difference between positive and negative emotions ($p > .05$), and there was none significant difference between the three emotions at the clear level (all p values $> .05$). The main effects of emotion and clarity were not significant, all p values $> .05$ (Fig. 2, Tables 2 and 3).

N170 The main effect of emotion was significant [F (2, 60) = 10.35, $p < .001$, $\eta_p^2 = .256$]. N170 was larger in negative than in neutral and positive emotions (all p values $< .05$). There was no significant difference between neutral and positive emotions ($p > .05$). The main effect of clarity was significant [F (1, 30) = 93.75, $p < .001$, $\eta_p^2 = .758$]. Clear stimuli led to a larger N170 than degraded stimuli. The interaction effect of clarity by emotion was significant [F (2, 60) = 5.17, $p = .009$, $\eta_p^2 = 0.147$]. At the clear level, N170 was larger in negative than in neutral and positive emotions ($p = .001$ and $p < .001$). There was no significant difference between neutral and positive emotions ($p > .05$). At the degraded level, the simple effect of emotion was not significant, $p > .05$ (Fig. 2, Tables 2 and 3).

EPN The main effect of emotion was significant [F (1.68, 50.27) = 30.64, $p < .001$, $\eta_p^2 = 0.505$]. EPN was larger in negative and positive emotions than in neutral emotions (all $p < .001$) and larger in positive than in negative emotions ($p < .001$). The main effect of clarity was significant [F (1, 30) = 1.28, $p < .001$, $\eta_p^2 = 0.391$]. Clear stimuli led to a larger EPN than degraded stimuli. The interaction effect of clarity by emotion was significant [F (2, 60) = 5.45, $p = .007$, $\eta_p^2 = 0.154$]. At the clear level, EPN was larger in positive than in negative and neutral emotions ($p = .001$ and $p < .001$), and there was no significant difference between negative and neutral emotions ($p = .001$ and $p < .001$). At the degraded level, EPN was larger in positive emotions than in negative and neutral emotions (all p values $< .05$) and a larger in negative than in neutral emotions, $p < .001$ (Fig. 2, Tables 2 and 3).

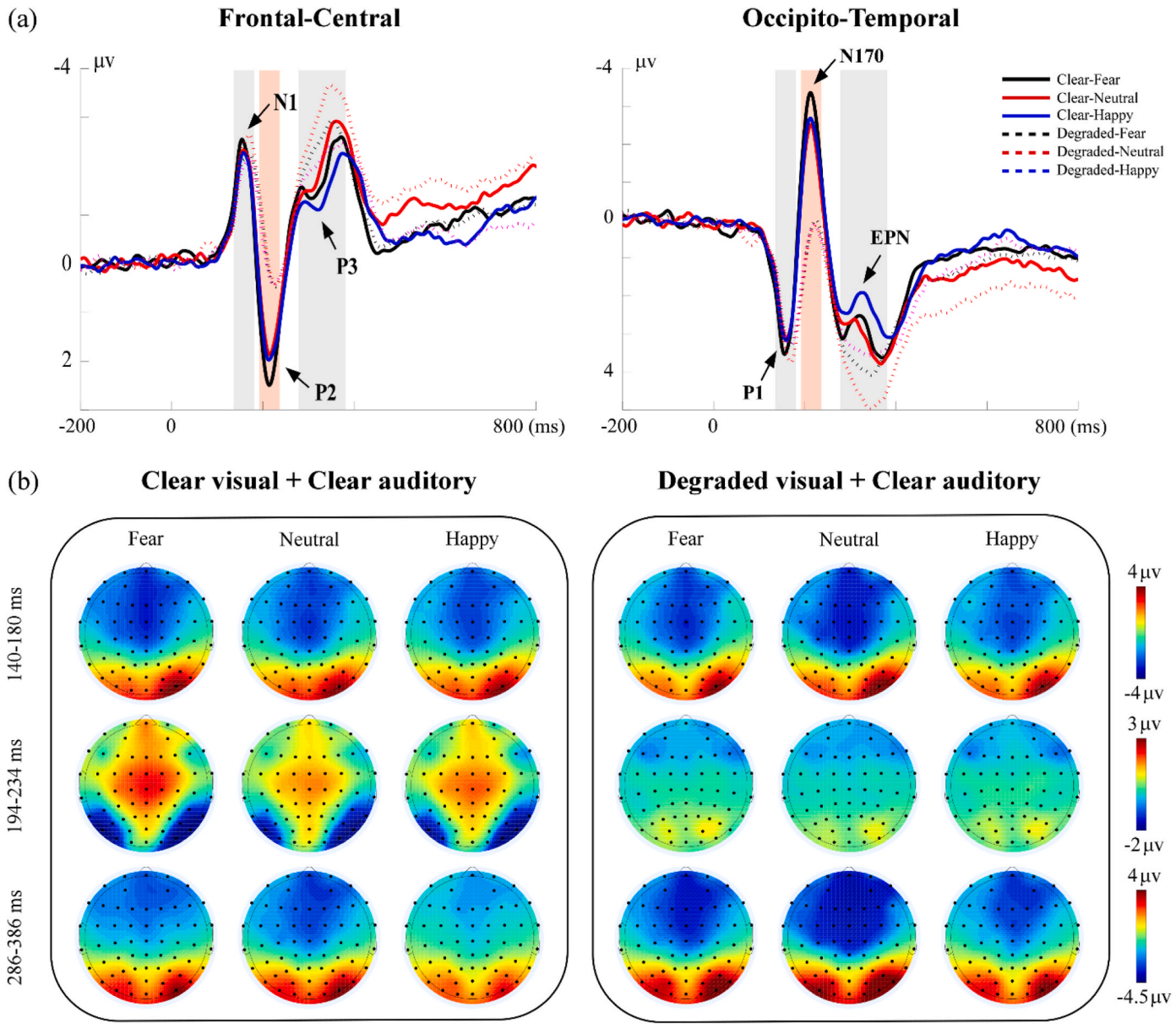


Fig. 2. **(a)** ERP waveforms were calculated by the mean average value of the selected electrodes from occipitotemporal electrode sites (P7/PO7/P8/PO8). There were significant differences in P1, N170, and EPN components induced by negative, neutral, and positive audiovisual integration in the occipitotemporal region. The black line represents negative emotions, the red line represents neutral emotions, and the blue line represents positive emotions. The full line represents the clear visual + clear auditory (AcVc) stimuli and the dashed line represents the clear auditory + degraded visual (AcVd) stimuli. **(b)** The topographic maps show the four ERP components of the six conditions, respectively.

Table 3
The main effect and interaction of each variable on the mean average value of ERP waveforms.

Component	Emotion			Clarity			Emotion * Clarity		
	F	*P*	η_p^2	*F*	*P*	η_p^2	*F*	*P*	η_p^2
N1	4.58	**.014**	.133	.41	.526	.014	2.86	.065	.087
P2	6.24	**.003**	.172	60.12	**.000**	.667	4.61	**.014**	.133
P3	24.80	**.000**	.453	13.37	**.001**	.292	2.05	.138	.064
P1	2.29	.111	.071	2.99	.094	.091	5.89	**.005**	.164
N170	10.35	**.000**	.256	93.75	**.000**	.758	5.17	**.009**	.147
EPN	30.64	**.000**	.505	19.28	**.000**	.391	5.45	**.007**	.154

3.3. Microstate result

The current study implemented cross-validation to determine the optimal number of microstate maps, resulting in the detection of five maps that accounted for over 85% of the variance in the test set. These maps were subsequently employed in further analyses. By fitting the microstate maps to the grand average ERP within a time frame of 100–800ms, the same five microstate maps and temporal sequence were

identified across six experimental conditions. The topographies of these microstate maps, as well as the chain of processes, are illustrated in Fig. 3. Table 4 provides details regarding the onset, offset, and duration of each microstate map. Notably, microstate 3 dominated the clear condition during the time of 194–244ms, whereas microstate 4 dominated the degraded condition. Additionally, a significant difference in the duration of microstate 1 between conditions was revealed by the randomized statistics of the microstate diagram parameters. Further, an interaction between clarity and emotion was observed. For more information, please refer to the appendix.

4. Discussion

The primary aim of this study is to gain a comprehensive understanding of how the reliability of visual emotions influences emotional AV perception. The investigation examined the effects of visual noise on neural activity, specifically focusing on temporal windows between 194-234ms and 286–386ms. These findings demonstrate an adaptable response to the interference caused by visual noise, shedding new light on the neural mechanisms underlying the processing of AV stimuli. Furthermore, the results emphasize the importance of visual clarity in the integration of sensory signals.

4.1. The behavioral evidence for emotional AV modulated by visual clarity

The current study revealed that the recognition of emotions from AcVd stimuli was more challenging, as indicated by the slower

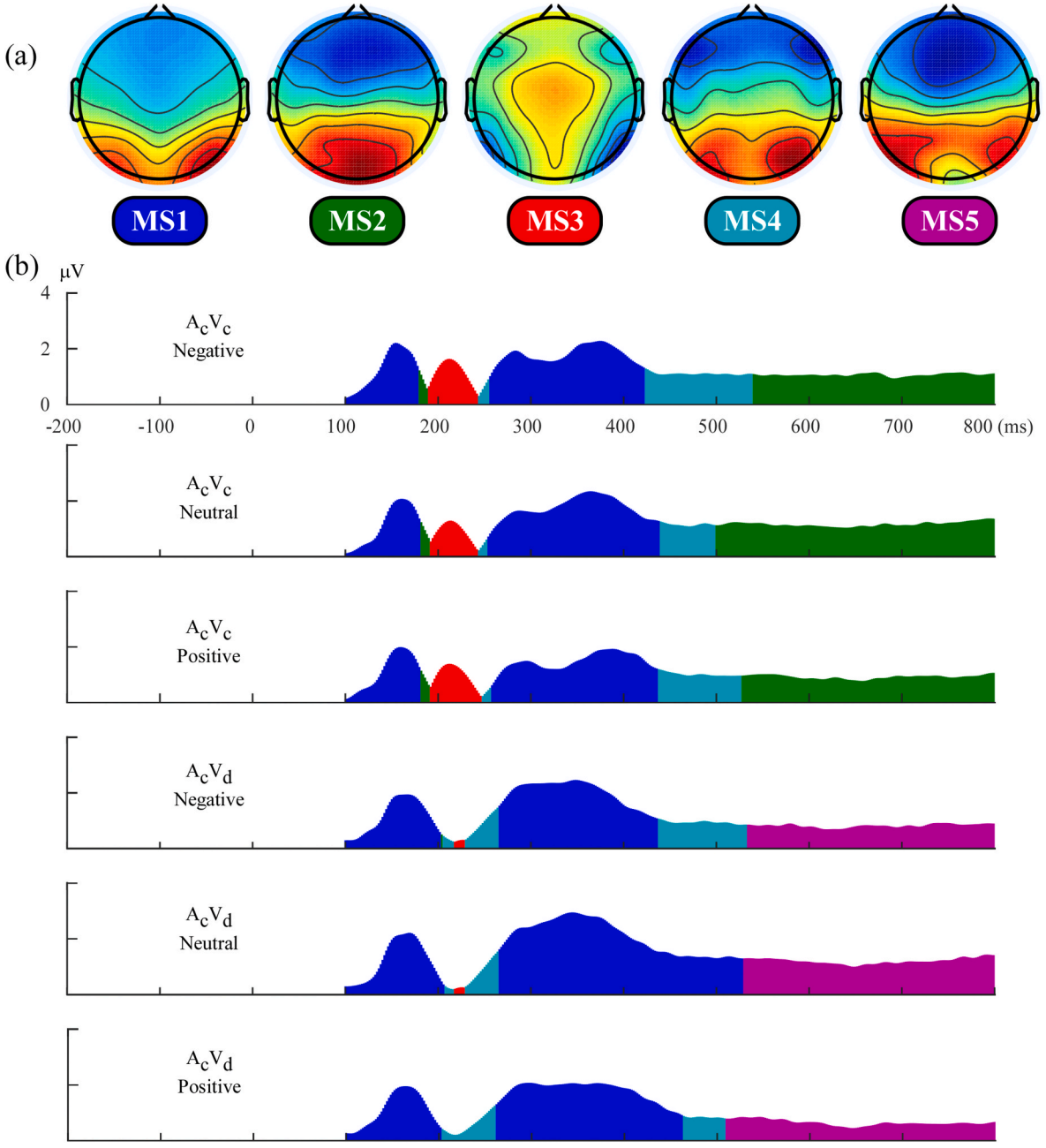


Fig. 3. Microstate result. **(a)** The optimal five microstate maps. All maps have been scaled to have a GFP of one. Different label colors are used to distinguish the microstate map and correspond to the colors in (b) (MS = microstate). Blue map areas indicate negative, red areas positive values. **(b)** The five-class solution of the microstate assignment for the 6 conditions. The y-axis represents the GFP value (in μV) and the x-axis represents time (in ms). Different colors represent specific microstate classes.

Table 4
Descriptive onset, offset, and duration (in ms) of microstate maps (analysis time window 100~800ms).

MS Class	Fitting parameter	AcVc			AcVd		
		Negative	Neutral	Positive	Negative	Neutral	Positive
MS1	Onset (ms)	102	102	102	102	102	102
	Offset (ms)	424	440	438	438	530	464
	Duration (ms)	248	268	262	276	372	306
MS2	Onset (ms)	182	184	184	206	Null	Null
	Offset (ms)	802	802	802	206	Null	Null
	Duration (ms)	272	312	284	2	Null	Null
MS3	Onset (ms)	192	194	194	220	220	Null
	Offset (ms)	244	244	248	230	230	Null
	Duration (ms)	54	52	56	12	12	Null
MS4	Onset (ms)	246	246	250	208	210	206
	Offset (ms)	540	500	528	534	266	510
	Duration (ms)	128	70	100	144	46	104
MS5	Onset (ms)	Null	Null	Null	536	532	512
	Offset (ms)	Null	Null	Null	802	802	802
	Duration (ms)	Null	Null	Null	268	272	292

processing and lower accuracy in emotional recognition compared to AcVc stimuli. These findings are consistent with previous research conducted by de Boer et al. (2021, b), which similarly reported decreased accuracy and increased cognitive effort when deciphering emotional stimuli in the presence of visual noise. The current findings highlight the importance of clear visual signals to improve emotional perception and understanding. Meanwhile, the individuals had a significantly higher level of accuracy for the emotional recognition of the AcVd stimuli than at a random level. It means that emotional sound could make a great enhancement to emotional recognition for visually impaired individuals, but it could not fully compensate for the negative effects of visual impairment (de Boer et al., 2021a,b). One research found that hearing aids can improve emotion recognition performance of older adults, but still worse than that of normal hearings individuals (Goy et al., 2016). It means that individuals rely mostly on visual information to complete the emotion perception (Collignon et al., 2008). However, these advantages were not absolute and fixed (Alais and Burr, 2004). Recently, Shaikh (2022) evaluated MSI processing based on the reliability and relative weighting of visual and auditory information. They want to verify the effect from noise on the development of modality-specific synaptic weights by a computational model. It found that the synaptic weight of visual cues was relatively larger than that of auditory, but the latter would be with larger synaptic weight when the visual cues was significantly degraded (Shaikh, 2022). The synaptic weights were dynamically developed with the different noise levels between the visual and auditory cues. Unlike the healthy participants in the current study, the visually degraded individuals could make a better adaptation to the visually degraded signals. And the experience may reduce their dependence on the visual modality. Therefore, it is possible to focus on individuals with audio-visual deficits caused by visual impairment (AcVd or AdVd) in further studies.

Participants exhibited high mean accuracy for both clear and degraded stimuli. However, how did participants utilize the visual and auditory cues to execute the task? Unfortunately, we cannot provide direct evidence to address this question in the current study, but we can propose one probable explanation through the MLE (Deneve and Pouget, 2004; Kersten and Yuille, 2003). The visual and auditory signals were integrated based on appropriate weighting to enhance perception's reliability and accuracy, and the weighting of vision and sound was to be flexible and dynamic (Shaikh, 2022). This viewpoint aligns with the findings of related research (Collignon et al., 2008). Collignon et al. (2008) found that, compared with AcVc stimuli, the AcVd stimuli made a stronger redundancy gain (Kayser et al., 2007). Moreover, they found that the participants oriented more toward visual modality when visual signals were clear and more toward auditory when visual signals were noise. These findings meant that the perceptual superiority of vision and hearing was not fixed or absolute (Alais and Burr, 2004). The next section will expound on the impact of different visual clarity levels on neural activity during the processing of AV stimuli and its associated changes in temporal dynamics.

4.2. The ERP components of emotional AV modulated by visual clarity

The neural processing of AV stimuli was explored through recording ERPs in our study. In terms of time course, we made the following discussion of the findings in 140–180ms, 194–234ms, and 286–386ms time windows, respectively.

Our findings indicated that emotional AV stimuli, whether negative or positive, elicited a smaller P1 amplitude compared to neutral stimuli. This suggested a rapid interpretation of emotional information and reduced attentional resource input (Vuilleumier and Pourtois, 2007). Additionally, perceptual priority was assigned to both negative (Neuberg et al., 2011) and positive (Carstensen and DeLiema, 2018) emotional stimuli when visual clarity was obscured by noise due to the attention capture effect, an automatic bottom-up attention regulation (Brosch et al., 2007, 2009). Interestingly, this attentional bias was only observed for AcVd stimuli, not AcVc stimuli, which we attributed to differences in the low-level visual stimuli characteristics. Previous research showed that P1 was sensitivity to physical features of stimuli (Pfütze et al., 2002). Rousselet et al. (2008) found that the sensitivity to the visual noise started about 120–130ms after stimulus onset and continued for 25–40ms (Rousselet et al., 2008). The current finding in the time range of the P1 component suggests that the sensitivity to the physical features of the emotional signals was significant in response to the degraded AV stimuli. Moreover, analysis of microstate 1 of the ERP waveforms revealed a similar processing pattern for AV stimuli, regardless of visual clarity. Our findings show that in the early processing stage of emotional AV stimuli, the clarity of visual stimuli and emotional valence jointly regulate the signal encoding processing, and the underlying neural activity pattern of emotional AV stimuli processing is the same.

Compared with the AcVd stimuli, the AcVc stimuli induced a larger N170 amplitude, suggesting that the N170 amplitude was inversely linear with the SNR of visual clarity. However, the earlier P1 was not affected by the above. Rousselet et al. (2008) suggested that the extraction of task-related information started at the transition between the P1 and the N170 and peaked circa at about N170 component. In the current study, from the ERP of AcVc, either in frontocentral or occipitotemporal clusters, there was an attentional bias toward negative emotions (Luo et al., 2010; Zhang and Luo, 2013). However, the attentional bias towards fearful emotions was not observed in the degraded visual condition, indicating that visual reliability is crucial for emotional AV perception (de Boer et al., 2021a; de Boer et al., 2021b). These findings proposed that the activity on the N170 component was

related to the extraction of task-related information, and the visual clarity significantly affected neural regulation mechanisms for emotional AV processing. Specifically, the physical characteristics of visual stimuli limit the function of N170 components as emotional attentional capture.

The identification of microstates provided compelling evidence for different neural regulatory mechanisms between the AcVd and AcVc conditions. During the time window of 194–234ms, Microstate 3 was associated with the AcVc condition, and microstate 4 was linked to the AcVd condition, indicating the existence of distinct processing mechanisms. Further analysis using source analysis demonstrated that, in comparison to the AcVd condition, the AcVc condition produced stronger neural responses in the lingual gyrus, the right fusiform gyrus (rFG), the middle temporal gyrus (MTG), and the middle occipital gyrus (MOG), as depicted in Fig. 4(a). Notably, MTG, similar to STS, has been shown to be involved in the processing of AV information (Calvert et al., 2000). Thus, the observed lower activity in MTG in response to AcVd stimuli may be due to degraded vision preventing proper integration with the auditory signal. This interpretation is supported by a study by Drijvers et al. (2018), where clear and degraded voices were coupled with consistent or inconsistent gestures (Drijvers et al., 2018). The authors reported that activity in the posterior superior temporal sulcus (p-STS), MTG, and medial temporal lobe was lower in response to degraded stimuli.

What did the activity during the 194–234ms window represent? Our previous study suggested that this time window primarily represents the processing of face structure in AV stimuli (Liang et al., 2022). In the present study, we investigated the relationship between the N170 and P2 components and the processing of AcVc stimuli, as well as the activation of the middle temporal gyrus (MTG), middle occipital gyrus (MOG), and fusiform gyrus (FG) during the perception of facial and vocal emotions. Analyses of both ERP and microstate revealed a significant effect of visual clarity on neural activity, consistent with the behavioral results observed. We provide some speculative explanations for the observed neural mechanisms involved in the re-weighting of visual reliability. Previously established research findings provide reliable evidence for our speculation that the MTG and MOG regions play a crucial role in visual weighting and are responsible for assessing the effectiveness of sensory signals. Moreover, the MOG, in particular, is known to be a primary visual processing area that plays a major role in visual weighting (Chechlacz et al., 2013). Our results suggest that the more clearly visual signals have more credibility, thus leading to stronger neural activity. On the other hand, our findings on MTG support the speculation by Hertz and Amedi (2015) that this region is sensitive to the physical properties of sensory input signals, such as noise (Hertz and Amedi, 2015). Combining the results of microstate and ERP amplitude analyses during 194–248ms revealed that visual clarity affected the pattern of neural activity while emotional information affected its intensity. We speculate that different pathways may be involved in the brain's processing of AV stimuli with different emotional valence and visual clarity.

The P3 and EPN findings indicated that ERP activation was more pronounced for negative and positive emotions compared to neutral emotions. These results align with previous research (Schupp et al., 2007, 2008), which suggests that AV stimuli with higher emotional arousal are evaluated as possessing survival significance, signifying a subjective motivation for preferential top-down regulation processing (Lee et al., 2010). Furthermore, microstate 1 demonstrated a significant interaction between clarity and emotion in the offset time and duration indicators. Specifically, under the degraded condition, the duration of neutral emotion was significantly longer compared to negative and positive emotions, indicating that the processing time required for

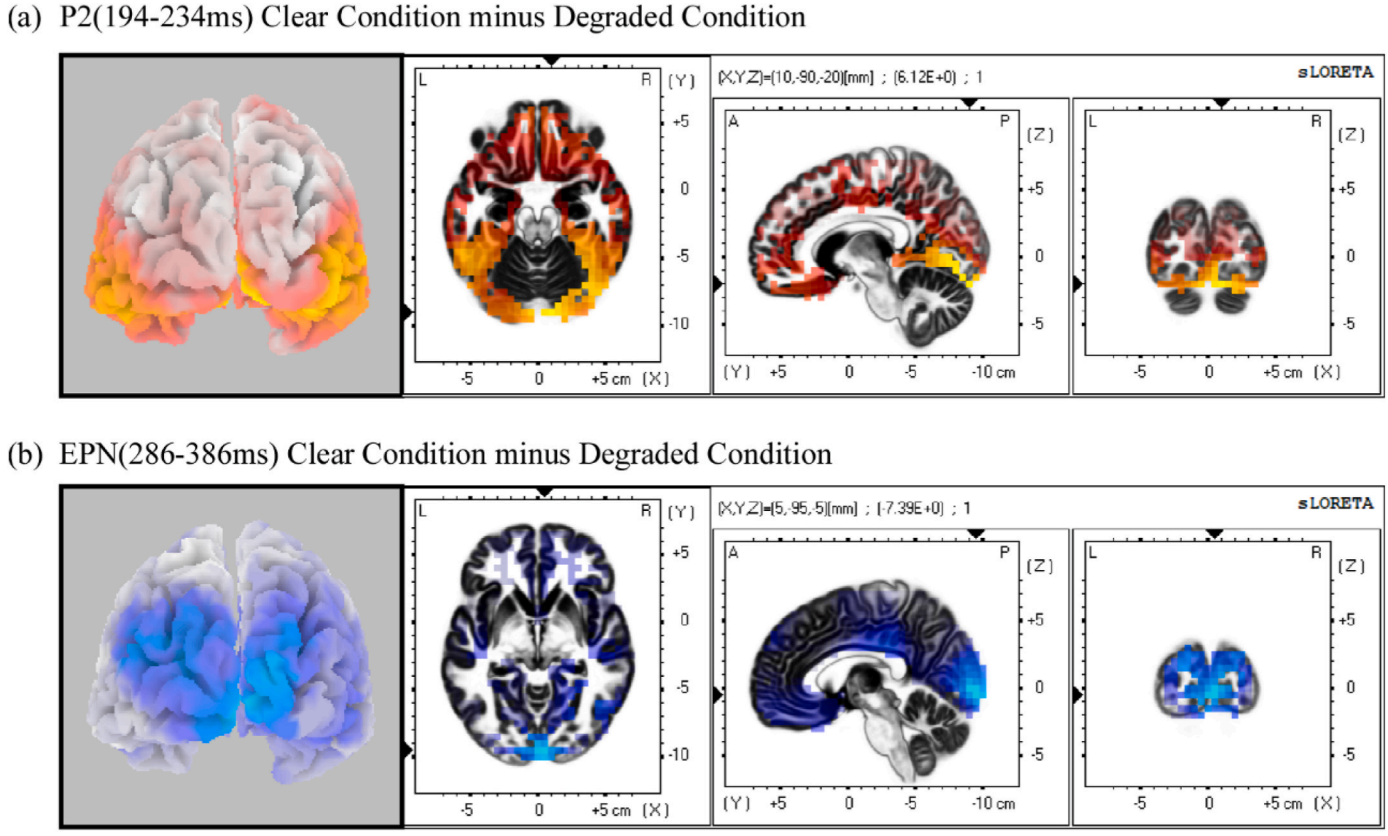


Fig. 4. **(a)** The lingual gyrus, right fusiform gyrus (rFG), middle temporal gyrus (MTG) and middle occipital gyrus (MOG) had stronger neural responses in the AcVc condition than in the AcVd condition [t (0.05) = 4.456, Extreme p = .0022]. **(b)** The lingual gyrus and cuneus had stronger neural responses in the AcVd condition than in the AcVc condition [t (0.05) = 4.583, Extreme p = .0002].

neutral information was longer compared to emotional information.

The results of both EPN and P300 were consistent with those of N170 and P2, indicating larger ERP components in response to AcVc stimuli. However, microstate analysis provided a more detailed insight into the time course of cognitive processing. Both the AcVc and AcVd conditions showed the presence of microstate 1 between 260 and 438ms. Furthermore, source analysis revealed stronger neural activity in the superior parietal lobule (SPL), precuneus, and cuneus. The IPS is formed by the SPL and the inferior parietal lobule (IPL), and it has strong connections with multiple brain regions such as the occipital and frontal motor cortices, thereby enabling it to conduct various functions like multimodal perception and attention allocation. The present findings suggest that the SPL exhibited top-to-down regulation in the perceptual processing of AV stimuli, whether AcVc or AcVd. Although these findings are not entirely consistent with Regenbogen et al. (2018), the differences will be discussed in detail later.

Microstate 1, associated with detailed emotional processing, showed a longer duration in the AcVd condition compared to the AcVc condition, indicating that processing AcVd stimuli required more time. To further investigate these differences, a source analysis was conducted. The results revealed stronger neural responses for AcVd stimuli in the lingual gyrus and cuneus compared to AcVc stimuli, which are known to be functionally relevant for the higher-order processing of visual signals (Jola et al., 2013). Specifically, these regions are responsible for multisensory association areas that process visual inputs (Vachon et al., 2013). These findings suggest that the degree of visual clarity influences processing in the primary visual cortex, with clearer visual cues resulting in stronger neural responses and lower dependence on the primary visual cortex (Stefanou et al., 2019). Conversely, AcVd stimuli required a stronger dependence on the visual cortex due to the greater effort needed to evaluate their meaningful signals. Overall, this study demonstrated that the potential regulatory mechanism of the EPN and P3 components was largely dependent on higher-order visual processing, specifically top-down regulation.

4.3. The effect of emotional information on signal accuracy rating

Further analysis was carried out to investigate the potential reasons for the observed differences between our study and the study conducted by Regenbogen et al. (2018). In their study, they aimed to examine whether brain regions modulate the processing of AV stimuli with varying degrees of visual clarity by using functional magnetic resonance imaging (fMRI). They reported that the neural responses for AdVd in the intraparietal sulcus (IPS) were stronger than those for AcVc. Conversely, the neural responses for AcVc in the lingual gyrus were stronger than those for AdVd. They postulated that direct information transmission between the visual and auditory cortices facilitated the processing of AcVc, whereas that of AdVd was realized through top-down mechanisms of the IPS. This process of regulation, which involves higher-order areas, seems to occur at a later stage, such as the EPN or P300 time window. However, our microstate analysis suggests that it may have occurred earlier.

Due to the low temporal precision of fMRI, we chose to utilize ERPs for their higher temporal resolution. The current study provides supplementary information on the time course of the DCM model. Specifically, the study discovered three main findings that illuminated the underlying neural mechanisms of AV integration. Firstly, activity in the lingual gyrus was stronger for AcVc than AcVd during the 194–234ms time window. Secondly, stronger neural activity was observed in the SPL (part of the IPL) during the 260–438ms time window for both AcVc and AcVd stimuli. Lastly, stronger neural responses for AcVd were found in the lingual gyrus and cuneus than for AcVc. These findings demonstrate the dynamic and flexible regulation of AV stimuli in the brain during specific time windows.

There appears to be a more specific time window for the top-down regulation of the intraparietal sulcus (IPS) for both clear auditory + clear visual (AcVc) and clear auditory + degraded visual (AcVd) stimuli. Between 194 and 234ms, the processing of AcVc stimuli primarily engaged the visual and auditory cortices. From 260 to 438ms, the visual cortex played a more prominent role in the processing of degraded auditory + degraded visual (AdVd) stimuli. These findings align with the "minimum effort principle" (Zipf, 1949), as top-down regulation was observed to reallocate cognitive resources and facilitate adequate processing of AcVd stimuli during the P3 and EPN stages (Briggs and Martin, 2009). In summary, the ERP components elicited by emotional AV stimuli serve distinct psychological functions and involve different brain regions at different time intervals. Although DCM is useful for predicting the dynamic trends of visual and auditory primary cortex processing as effectiveness changes, it is insufficient for fully explaining the neural mechanisms of AV integration. This study's findings serve as a complement to current research models, shedding light on the underlying processing mechanism of emotional AV, which involves the modulation of the connection between the primary visual and auditory cortex by higher order regions. This modulation is associated with the level of reliance on the visual cortex, which is influenced by visual clarity. Ultimately, the results reflect the flexible regulatory mechanism of higher brain regions on primary cortex processing.

5. Limitations and future directions

The primary aim of this study was to investigate the mechanisms underlying the impact of impaired vision on audiovisual (AV) processing across different stages. ERP techniques were utilized for this purpose. However, it is important to acknowledge the limitations of EEG source localization in accurately determining brain activation, which may affect the interpretation of the results. Secondly, the study did not collect post-experiment reports from participants, making it challenging to determine the sensory modality utilized in their decision-making process. Future studies should consider incorporating subjective data (i.e., the possibility that participants may make decisions based on visual or auditory modality dominance for the AV stimuli in the current study), objective decision-making behaviors, and EEG data to establish correlations and develop a comprehensive theoretical model for emotional AV integration. In addition, six degraded levels of visual signal were presented, and averaging them for the behavioral and EEG data interfere with some findings.

6. Conclusion

The current study aimed to investigate the impact of the clarity of visual signals on the emotional perception of AV stimuli, with a specific focus on two crucial stages. The analysis of N170 and P2 demonstrated that clear negative facial expressions accelerated primary processing. Conversely, degraded visual signals impeded primary processing because of visual noise. These findings indicate that emotional content and physical characteristics are distinct entities but interact in complex ways. Regarding EPN and P3, comparing AcVc and AcVd illustrated that the latter was successful due to top-down attention regulation during the perceptual encoding stage. Overall, the results suggest that both AcVc and AcVd activation are related to both physical vision characteristics and emotional arousal. Thus, it is essential to consider both factors in future studies. These findings contribute to our understanding of the neural mechanisms involved in processing emotional and AV stimuli with distinct degrees of clarity and emotional valence.

CRediT authorship contribution statement

Yuchen Li: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing - original draft, Writing - review & editing. **Jing Wang:** Investigation, Writing - review & editing. **Junyu Liang:** Investigation, Writing - review & editing. **Chuanlin Zhu:** Conceptualization, Supervision. **Zhao Zhang:**

Conceptualization, Project administration, Resources, Supervision. **Wenbo Luo:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.neuropsychologia.2023.108785.

References

Alais, D., Burr, D., 2004. The ventriloquist effect results from near-optimal bimodal integration. Curr. Biol. 14 (3), 257–262. https://doi.org/10.1016/j.cub.2004.01.029.

Balconi, M., Carrera, A., 2011. Cross-modal integration of emotional face and voice in congruous and incongruous pairs: the P2 ERP effect. J. Cognit. Psychol. 23 (1), 132–139. https://doi.org/10.1080/20445911.2011.473560.

Belin, P., Fillion-Bilodeau, S., Gosselin, F., 2008. The Montreal Affective Voices: a validated set of nonverbal affect bursts for research on auditory affective processing. Behav. Res. Methods 40 (2), 531–539. https://doi.org/10.3758/BRM.40.2.531.

Bhat, J., Miller, L.M., Pitt, M.A., Shahin, A.J., 2015. Putative mechanisms mediating tolerance for audiovisual stimulus onset asynchrony. J. Neurophysiol. 113 (5), 1437–1450. https://doi.org/10.1152/jn.00200.2014.

Blanco-Elorrieta, E., Ding, N., Pylkkanen, L., Poeppel, D., 2020. Understanding requires tracking: noise and knowledge interact in bilingual comprehension. J. Cognit. Neurosci. 32 (10), 1975–1983. https://doi.org/10.1162/jocn_a_01610.

Briggs, K.E., Martin, F.H., 2009. Affective picture processing and motivational relevance: arousal and valence effects on ERPs in an oddball task. Int. J. Psychophysiol. 72 (3), 299–306. https://doi.org/10.1016/j.ijpsycho.2009.01.009.

Brosch, T., Sander, D., Scherer, K.R., 2007. That baby caught my eye... attention capture by infant faces. Emotion 7 (3), 685–689. https://doi.org/10.1037/1528-3542.7.3.685.

Brosch, T., Grandjean, D., Sander, D., Scherer, K.R., 2009. Cross-modal emotional attention: emotional voices modulate early stages of visual processing. J. Cognit. Neurosci. 21 (9), 1670–1679. https://doi.org/10.1162/jocn.2009.21110.

Calvert, G.A., Campbell, R., Brammer, M.J., 2000. Evidence from functional magnetic resonance imaging of crossmodal binding in the human heteromodal cortex. Curr. Biol. 10 (11), 649–657. https://doi.org/10.1016/s0960-9822(00)00513-3.

Carstensen, L.L., DeLiema, M., 2018. The positivity effect: a negativity bias in youth fades with age. Curr. Opin. Behav. Sci. 19, 7–12. https://doi.org/10.1016/j.cobeha.2017.07.009.

Chai, Y., Liu, T.T., Marrett, S., Li, L., Khojandi, A., Handwerker, D.A., Bandettini, P.A., 2021. Topographical and laminar distribution of audiovisual processing within human planum temporale. Prog. Neurobiol. 205, 102121 https://doi.org/10.1016/j.pneurobio.2021.102121.

Chechlacz, M., Terry, A., Demeyere, N., Douis, H., Bickerton, W.L., Rotshtein, P., Humphreys, G.W., 2013. Common and distinct neural mechanisms of visual and tactile extinction: a large scale VBM study in sub-acute stroke. Neuroimage Clin. 2, 291–302. https://doi.org/10.1016/j.nicl.2013.01.013.

Cohen, J., 1992. Statistical power analysis. Curr. Dir. Psychol. Sci. 1 (3), 98–101. https://doi.org/10.1111/1467-8721.ep10768783.

Colijn, J.M., Buitendijk, G.H.S., Prokofyeva, E., Alves, D., Cachulo, M.L., Khawaja, A.P., European Eye Epidemiology, c, 2017. Prevalence of age-related macular degeneration in europe: the past and the future. Ophthalmology 124 (12), 1753–1763. https://doi.org/10.1016/j.ophtha.2017.05.035.

Collignon, O., Girard, S., Gosselin, F., Roy, S., Saint-Amour, D., Lassonde, M., Lepore, F., 2008. Audio-visual integration of emotion expression. Brain Res. 1242, 126–135. https://doi.org/10.1016/j.brainres.2008.04.023.

de Boer, M.J., Jurgens, T., Baskent, D., Cornelissen, F.W., 2021a. Auditory and visual integration for emotion recognition and compensation for degraded signals are preserved with age. Trends Hear. 25 https://doi.org/10.1177/23312165211045306, 23312165211045306.

de Boer, M.J., Jurgens, T., Cornelissen, F.W., Baskent, D., 2021b. Degraded visual and auditory input individually impair audiovisual emotion recognition from speech-like stimuli, but no evidence for an exacerbated effect from combined degradation. Vis. Res. 180, 51–62. https://doi.org/10.1016/j.visres.2020.12.002.

De Gelder, B., Vroomen, J., 2000. The perception of emotions by ear and by eye. Cognit. Emot. 14 (3), 289–311. https://doi.org/10.1080/026999300378824.

Deneve, S., Pouget, A., 2004. Bayesian multisensory integration and cross-modal spatial links. J. Physiol. Paris 98 (1–3), 249–258. https://doi.org/10.1016/j.jphysparis.2004.03.011.

Doi, H., Shinohara, K., 2015. Unconscious presentation of fearful face modulates electrophysiological responses to emotional prosody. Cerebr. Cortex 25 (3), 817–832. https://doi.org/10.1093/cercor/bht282.

Drijvers, L., Ozyurek, A., Jensen, O., 2018. Alpha and beta oscillations index semantic congruency between speech and gestures in clear and degraded speech. J. Cognit. Neurosci. 30 (8), 1086–1097. https://doi.org/10.1162/jocn_a_01301.

Ernst, M.O., Bulthoff, H.H., 2004. Merging the senses into a robust percept. Trends Cognit. Sci. 8 (4), 162–169. https://doi.org/10.1016/j.tics.2004.02.002.

Frühholz, S., Jellinghaus, A., Herrmann, M., 2011. Time course of implicit processing and explicit processing of emotional faces and emotional words. Biol. Psychol. 87 (2), 265–274. https://doi.org/10.1016/j.biopsycho.2011.03.008.

Gong, X., Huang, Y.-X., Wang, Y., Luo, Y.-J., 2011. Revision of the Chinese facial affective picture system. Chin. Ment. Health 25 (1), 40–46. https://psycnet.apa.org/record/2011-05085-005.

Goy, H., Pichora-Fuller, M.K., Singh, G., Russo, F.A., 2016. Perception of emotional speech by listeners with hearing aids. Can. Acoust. 44 (3) jcaa.caa-aca.ca/index.php/jcaa/article/view/2962.

Habermann, M., Weusmann, D., Stein, M., Koenig, T.J. F.i. n., 2018. A students guide to randomization statistics of multichannel event related potentials using Ragu 12, 355. https://doi.org/10.3389/fnins.2018.00355.

Hertz, U., Amedi, A., 2015. Flexibility and stability in sensory processing revealed using visual-to-auditory sensory substitution. Cerebr. Cortex 25 (8), 2049–2064. https://doi.org/10.1093/cercor/bhu010.

Johnson, A.P., Woods-Fry, H., Wittich, W., 2017. Effects of magnification on emotion perception in patients with age-related macular degeneration. Investigative Ophthalmology and Visual Science 58 (5), 2520–2526. https://doi.org/10.1167/iovs.16-21349.

Jola, C., McAleer, P., Grosbras, M.H., Love, S.A., Morison, G., Pollick, F.E., 2013. Uni- and multisensory brain areas are synchronised across spectators when watching unedited dance recordings. Iperception 4 (4), 265–284. https://doi.org/10.1068/i0536.

Kayser, C., Petkov, C.I., Augath, M., Logothetis, N.K., 2007. Functional imaging reveals visual modulation of specific fields in auditory cortex. J. Neurosci. 27 (8), 1824–1835. https://doi.org/10.1523/JNEUROSCI.4737-06.2007.

Kersten, D., Yuille, A., 2003. Bayesian models of object perception. Curr. Opin. Neurobiol. 13 (2), 150–158. https://www.ncbi.nlm.nih.gov/pubmed/12744967.

Koenig, T., Melie-García, L.J. B.t., 2010. A method to determine the presence of averaged event-related fields using randomization tests 23 (3), 233–242. https://doi.org/10.1007/s10548-010-0142-1.

Koenig, T., Stein, M., Grieder, M., Kottlow, M.J. B.t., 2014. A tutorial on data-driven methods for statistically assessing ERP topographies 27 (1), 72–83. https://doi.org/10.1007/s10548-013-0310-1.

Lane, J., Rohan, E.M.F., Sabeti, F., Essex, R.W., Maddess, T., Dawel, A., McKone, E., 2018. Impacts of impaired face perception on social interactions and quality of life in age-related macular degeneration: a qualitative study and new community resources. PLoS One 13 (12), e0209218. https://doi.org/10.1371/journal.pone.0209218.

Lee, K.Y., Lee, T.H., Yoon, S.J., Yang, S.C., Choi, J.S., Kim, H.T., 2010. Neural correlates of top–down processing in emotion perception: an ERP study of emotional faces in white noise versus noise-alone stimuli. Brain Res. 1337 (none), 56–63. https://doi.org/10.1016/j.brainres.2010.03.094.

Lehmann, D., Ozaki, H., Pal, I.J.E., neurophysiology, c., 1987. EEG alpha map series: brain micro-states by space-oriented adaptive segmentation 67 (3), 271–288. https://doi.org/10.1016/0013-4694(87)90025-3.

Liang, J., Li, Y., Zhang, Z., Luo, W., 2022. Sound gaps boost emotional audiovisual integration independent of attention: evidence from an ERP study. Biol. Psychol. 168, 108246 https://doi.org/10.1016/j.biopsycho.2021.108246.

Luo, W., Feng, W., He, W., Wang, N.-Y., Luo, Y.-J., 2010. Three stages of facial expression processing: ERP study with rapid serial visual presentation 49 (2), 1857–1867. https://doi.org/10.1016/j.neuroimage.2009.09.018.

Murray, M.M., Brunet, D., Michel, C.M. J.B.t., 2008. Topographic ERP analyses: a step-by-step tutorial review. Brain Topogr. 20 (4), 249–264. https://doi.org/10.1007/s10548-008-0054-5.

Neuberg, S.L., Kenrick, D.T., Schaller, M., 2011. Human threat management systems: self-protection and disease avoidance. Neurosci. Biobehav. Rev. 35 (4), 1042–1051. https://doi.org/10.1016/j.neubiorev.2010.08.011.

Pfütze, E.M., Sommer, W., Schweinberger, S.R., 2002. Age-related slowing in face and name recognition: evidence from event-related brain potentials. Psychology and aging 17 (1), 140.

Pourtois, G., De Gelder, B., Vroomen, J., Rossion, B., Crommelinck, M., 2000. The time-course of intermodal binding between seeing and hearing affective information. Neuroreport 11 (6), 1329–1333. https://doi.org/10.1097/00001756-200004270-00036.

Regenbogen, C., Seubert, J., Johansson, E., Finkelmeyer, A., Andersson, P., Lundstrom, J. N., 2018. The intraparietal sulcus governs multisensory integration of audiovisual information based on task difficulty. Hum. Brain Mapp. 39 (3), 1313–1326. https://doi.org/10.1002/hbm.23918.

Romei, V., Murray, M.M., Merabet, L.B., Thut, G., 2007. Occipital transcranial magnetic stimulation has opposing effects on visual and auditory stimulus detection: implications for multisensory interactions. J. Neurosci. 27 (43), 11465–11472. https://doi.org/10.1523/JNEUROSCI.2827-07.2007.

Rousselet, G.A., Pernet, C.R., Sekuler, B.A.B., 2008. Parametric study of EEG sensitivity to phase noise during face processing. BMC Neurosci. https://doi.org/10.1186/1471-2202-9-98.

Schindler, S., Bublatzky, F., 2020. Attention and emotion: an integrative review of emotional face processing as a function of attention. Cortex 130, 362–386. https://doi.org/10.1016/j.cortex.2020.06.010.

Schupp, H.T., Stockburger, J., Bublatzky, F., Junghofer, M., Weike, A.I., Hamm, A.O., 2007. Explicit attention interferes with selective emotion processing in human extrastriate cortex. BMC Neurosci. 8, 16. https://doi.org/10.1186/1471-2202-8-16.

Schupp, H.T., Stockburger, J., Schmalzle, R., Bublatzky, F., Weike, A.I., Hamm, A.O., 2008. Visual noise effects on emotion perception: brain potentials and stimulus identification. Neuroreport 19 (2), 167–171. https://doi.org/10.1097/WNR.0b013e3282f4aa42.

Shaikh, D., 2022. Learning multisensory cue integration: a computational model of crossmodal synaptic plasticity enables reliability-based cue weighting by capturing stimulus statistics. Front. Neural Circ. 16, 921453 https://doi.org/10.3389/fncir.2022.921453.

Stefanou, M.E., Dundon, N.M., Bestelmeyer, P.E.G., Koldewyn, K., Saville, C.W.N., Fleischhaker, C., Klein, C., 2019. Electro-cortical correlates of multisensory integration using ecologically valid emotional stimuli: differential effects for fear and disgust. Biol. Psychol. 142, 132–139. https://doi.org/10.1016/j.biopsycho.2019.01.011.

Stein, B.E., Meredith, M.A., Wallace, M.T., 1993. The visually responsive neuron and beyond: multisensory integration in cat and monkey. Prog. Brain Res. 95, 79–90. https://www.ncbi.nlm.nih.gov/pubmed/8493355.

Vachon, P., Voss, P., Lassonde, M., Leroux, J.M., Mensour, B., Beaudoin, G., Lepore, F., 2013. Reorganization of the auditory, visual and multimodal areas in early deaf individuals. Neuroscience 245, 50–60. https://doi.org/10.1016/j.neuroscience.2013.04.004.

Vuilleumier, P., Pourtois, G., 2007. Distributed and interactive brain mechanisms during emotion face perception: evidence from functional neuroimaging. Neuropsychologia 45 (1), 174–194. https://doi.org/10.1016/j.neuropsychologia.2006.06.003.

White, B.E., Langdon, C., 2021. The cortical organization of listening effort: new insight from functional near-infrared spectroscopy. Neuroimage 240, 118324. https://doi.org/10.1016/j.neuroimage.2021.118324.

Yao, D., 2001. A method to standardize a reference of scalp EEG recordings to a point at infinity. Physiol. Meas. 22 (4), 693. https://doi.org/10.1088/0967-3334/22/4/305.

Zhang, Luo, Luo, 2013. Single-trial ERP analysis reveals facial expression category in a three-stage scheme. Brain Res. 1512, 78–88. https://doi.org/10.1016/j.brainres.2013.03.044.

Zipf, G., 1949. Human Behaviour and the Principle of Least-Effort. Reading: Addison-Wesley, Cambridge MA. https://doi.org/10.2307/2226729.